exist. This was the only way I could see to help Carl. So upon reaching Pasadena, I wrote Carl a long letter, explaining why his heroine could not use black holes for rapid interstellar travel, and suggesting that she use wormholes instead, and that somebody in the novel discover that exotic material can really exist and can be used to hold the wormholes open. Carl accepted my suggestion with pleasure and incorporated it into the final version of his novel, *Contact.*[4]

It occurred to me, after offering Carl Sagan my comments, that his novel could serve as a pedagogical tool for students studying general relativity. As an aid for such students, during the autumn of 1985 Mike Morris (one of my own students) and I began to write a paper on the general relativistic equations for wormholes supported by exotic material, and those equations' connection to Sagan's novel.

We wrote slowly. Other projects were more urgent and got higher priority. By the winter of 1987–88, we had submitted our paper to the *American Journal of Physics*, but it was not yet published; and Morris, nearing the end of his Ph.D. training, was applying for postdoctoral positions. With his applications, Morris enclosed the manuscript of our paper. Don Page (a professor at Pennsylvania State University and a former student of mine and Hawking's) received the application, read our manuscript, and fired off a letter to Morris.

"Dear Mike, . . . it follows immediately from Proposition 9.2.8 of the book by Hawking- & Ellis, plus the Einstein field equations, that *any* wormhole [requires exotic material to hold it open] . . . Sincerely, Don N. Page."

How stupid I felt. I had never studied global methods[5] (the topic of the Hawking and Ellis book) in any depth, and I was now paying the price. I had deduced on Interstate 5, with modest labor, that to hold a precisely spherical wormhole open one must thread it with exotic material. However, now, using global methods and with even less labor, Page had deduced that to hold *any* wormhole open (a spherical wormhole, a cubical wormhole, a wormhole with random deformations), one must thread it with exotic material. I later learned that Dennis Gannon and C. W. Lee reached almost the same conclusion in 1975.